UNIQUENESS AND EXACTNESS OF SERRE FUNCTORS

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Yoneda lemma. categories-lemma-yoneda (cross-chapter, named). Identifies representing objects up to unique isomorphism.
- Representable homological functors. derived-lemma-representable-homological (cross-chapter, named). Exactness of the dual Hom sequences.
- Matress formalism. remark-matress, remark-dual-matress (same chapter, not carded). Constructs the comparison arrow phi between the two triangles.

Source reference. Original reference 001P: focused-seeds/geometry/S9-eqv-Serre-functor/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 0FY5: sources/stacks-equiv/pinned-original-source.tex, source lines 378–378. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0149: focused-seeds/geometry/S9-eqv-Serre-functor/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05QB: focused-seeds/geometry/S9-eqv-Serre-functor/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 0H8C: sources/stacks-equiv/pinned-original-source.tex, source lines 421–421. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 0H8B: sources/stacks-equiv/pinned-original-source.tex, source lines 390–390. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is

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carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In the situation of Definition 0FY5. If a Serre functor exists, then it is unique up to unique isomorphism and it is an exact functor of triangulated categories.*

Proof. Given a Serre functor S the object $S(X)$ represents the functor $Y \mapsto \mathrm{Hom}_{\mathcal{T}}(X, Y)^\vee$. Thus the object $S(X)$ together with the functorial identification $\mathrm{Hom}_{\mathcal{T}}(X, Y)^\vee = \mathrm{Hom}_{\mathcal{T}}(Y, S(X))$ is determined up to unique isomorphism by the Yoneda lemma (Categories, Lemma 001P). Moreover, for $\varphi : X \to X'$, the arrow $S(\varphi) : S(X) \to S(X')$ is uniquely determined by the corresponding transformation of functors $\mathrm{Hom}_{\mathcal{T}}(X, -)^\vee \to \mathrm{Hom}_{\mathcal{T}}(X', -)^\vee$.

For objects X, Y of $\mathcal{T}$ we have

$$\begin{aligned} \mathrm{Hom}(Y, S(X)[1])^\vee &= \mathrm{Hom}(Y[-1], S(X))^\vee \\ &= \mathrm{Hom}(X, Y[-1]) \\ &= \mathrm{Hom}(X[1], Y) \\ &= \mathrm{Hom}(Y, S(X[1]))^\vee \end{aligned}$$

By the Yoneda lemma we conclude that there is a unique isomorphism $S(X[1]) \to S(X)[1]$ inducing the isomorphism from top left to bottom right. Since each of the isomorphisms above is functorial in both X and Y we find that this defines an isomorphism of functors $S \circ [1] \to [1] \circ S$.

Let (A, B, C, f, g, h) be a distinguished triangle in $\mathcal{T}$. We have to show that the triangle $(S(A), S(B), S(C), S(f), S(g), S(h))$ is distinguished. Here we use the canonical isomorphism $S(A[1]) \to S(A)[1]$ constructed above to identify the target $S(A[1])$ of $S(h)$ with $S(A)[1]$. We first observe that for any X in $\mathcal{T}$ the triangle $(S(A), S(B), S(C), S(f), S(g), S(h))$ induces a long exact sequence

$$\ldots \to \mathrm{Hom}(X, S(A)) \to \mathrm{Hom}(X, S(B)) \to \mathrm{Hom}(X, S(C)) \to \mathrm{Hom}(X, S(A)[1]) \to \ldots$$

of finite dimensional k-vector spaces. Namely, this sequence is k-linear dual of the sequence

$$\ldots \leftarrow \mathrm{Hom}(A, X) \leftarrow \mathrm{Hom}(B, X) \leftarrow \mathrm{Hom}(C, X) \leftarrow \mathrm{Hom}(A[1], X) \leftarrow \ldots$$

which is exact by Derived Categories, Lemma 0149. Next, we choose a distinguished triangle $(S(A), E, S(C), i, p, S(h))$ which is possible by axioms TR1 and TR2. We want to construct the dotted arrow making following diagram commute

$$\begin{array}{ccccccccc} S(C)[-1] & \xrightarrow{S(h[-1])} & S(A) & \xrightarrow{S(f)} & S(B) & \xrightarrow{S(g)} & S(C) & \xrightarrow{S(h)} & S(A)[1] \\ \| & & \| & & \uparrow{\scriptstyle \varphi} & & \| & & \| \\ S(C)[-1] & \xrightarrow{S(h[-1])} & S(A) & \xrightarrow{i} & E & \xrightarrow{p} & S(C) & \xrightarrow{S(h)} & S(A)[1] \end{array}$$

Namely, if we have φ, then we claim for any X the resulting map $\mathrm{Hom}(X, E) \to \mathrm{Hom}(X, S(B))$ will be an isomorphism of k-vector spaces. Namely, we will obtain

a commutative diagram

$$\begin{array}{ccccccccc}
\mathrm{Hom}(X, S(C)[-1]) & \longrightarrow & \mathrm{Hom}(X, S(A)) & \longrightarrow & \mathrm{Hom}(X, S(B)) & \longrightarrow & \mathrm{Hom}(X, S(C)) & \longrightarrow & \mathrm{Hom}(X, S(A)[1]) \\
\| & & \| & & \uparrow{\scriptstyle \varphi} & & \| & & \| \\
\mathrm{Hom}(X, S(C)[-1]) & \longrightarrow & \mathrm{Hom}(X, S(A)) & \longrightarrow & \mathrm{Hom}(X, E) & \longrightarrow & \mathrm{Hom}(X, S(C)) & \longrightarrow & \mathrm{Hom}(X, S(A)[1])
\end{array}$$

with exact rows (see above) and we can apply the 5 lemma (Homology, Lemma 05QB) to see that the middle arrow is an isomorphism. By the Yoneda lemma we conclude that φ is an isomorphism.

To find φ, let us form the matress of Remark 0H8B from the distinguished triangles $A \to B \to C \to A[1]$ and $S(A) \to E \to S(C) \to S(A)[1]$. By Remark 0H8C the k-linear dual of this is also a matress. Using that S is a Serre functor, we see as above that this dual matress is the double complex made from the triangles (the second of which we do not yet know to be distinguished) $S(A) \to E \to S(C) \to S(A)[1]$ and $S(A) \to S(B) \to S(C) \to S(A)[1]$. But the matress condition exactly tells us that the maps $S(A) \to S(A)$ and $S(C) \to S(C)$ fit into a morphism of triangles, i.e., that we have a φ making the diagram above commute. $\square$